

Sep 27 — 5 papers

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RoboTalk: Learning Multi-Robot Communication and Coordination from Multimodal Demonstrations

SKIM

Dorian Benhamou Goldfajn et al.

They generated about eight thousand synthetic kitchen-robot demonstrations in which a leader robot plans and talks to a follower in natural language, and showed that fine-tuning small vision-language models on that data raises success on unseen tasks from about two percent to seventy-seven percent.

PROBLEM Small on-device vision-language models have not been trained to learn inter-robot messages and skill selection together from demonstrations, especially in decentralized manipulation where each robot sees only part of the scene. Most prior multi-robot language work either uses large models with prompting or treats communication and action as separate problems.

RESULTS Fine-tuned models reach 77 percent success on held-out novel tasks, compared with roughly 2 percent for the same models without tuning. The abstract names no specific model sizes, no simulator, and no baseline other than the untuned models.

METHOD They built a synthetic pipeline that produces 7,950 multimodal trajectories across 53 mobile-manipulation kitchen tasks. Each trajectory follows a fixed leader-follower planning protocol and includes tool calls for perception, manipulation, navigation and messaging, plus rationale traces and paraphrased natural-language messages. Open-source small models are then fine-tuned to output both the messages and the skill-level tool calls.

CAVEAT The only baseline is an untuned model that has never seen the tool-call format, so most of the jump from 2 to 77 percent may come from learning the format and protocol rather than learning to coordinate. The abstract reports no ablation that removes communication or runs under partial observability without messages, so we cannot tell how much of the success actually depends on the messages. The leader-follower protocol is scripted, so nothing is negotiated, no conflict is resolved, and nothing is tested under lost or delayed messages. Held-out tasks come from the same synthetic kitchen domain, so novel here probably means recombined skills, not a real shift in distribution.

WHY IT MATTERS This is peripheral to your work. It is about kitchen manipulation with a fixed hierarchy, not about peers negotiating right of way, proposals and acceptances, or robustness when messages drop. The one transferable idea is the recipe of generating synthetic demonstrations with explicit message tool calls to train a small model that both communicates and acts. That could matter if you ever want a learned language-level intent channel between vehicles, but nothing here tests safety, timing or partial participation.

arXiv:2609.23997v1 PDF incremental

Bearing-Only Formation Tracking Control for Euler-Lagrange Multi-Agent Systems Without Inter-Agent Communication

SKIM

Zilong Song, Lu Liu, Gang Feng

A controller that keeps robots in a formation using only bearing measurements to their neighbours, with no radio link at all, while the leaders move at changing speeds.

PROBLEM Earlier bearing-only formation controllers without communication could only hold a formation that stays still. When leaders move at time-varying speeds, followers cannot see or estimate the leaders' velocity, because there is no channel to run a distributed observer on.

RESULTS The guarantee is theoretical and local. Velocity error goes to zero and position error stays bounded but does not vanish, and asymptotic tracking holds only when the leaders move at constant velocity. The abstract gives no numbers and no baselines, and the only evidence beyond the proofs is an unquantified simulation.

METHOD The key observation is that the rate of change of a bearing contains the relative velocity projected onto the bearing direction. Combined with bearing rigidity, this gives a damping term that stands in for the missing velocity error. That term feeds an auxiliary variable, which acts as a surrogate velocity error for an adaptive controller on Euler-Lagrange dynamics. The authors also define a set of initial conditions that preserves rigidity, prove the set stays invariant, and use a Filippov-based Lyapunov analysis to prove convergence.

CAVEAT The result is local and practical, not exact: it depends on starting inside a rigidity-preserving set, and position error only ends up bounded whenever leader velocity changes. The work is simulation only, and it ignores bearing sensing noise, occlusion, and limited field of view, which are exactly what break bearing-rate differentiation on real hardware.

WHY IT MATTERS This is peripheral to your work. It removes communication on purpose, while your focus is explicit negotiation and intent exchange between vehicles. The one weak link is the fallback case where messages are lost and vehicles can only sense each other, but a generic Euler-Lagrange robot formation with no road geometry, no lane structure and no string stability analysis will not carry over directly to platooning.

arXiv:2609.29049v1 PDF incremental

Exact Average Consensus under Noisy Communication Links: A Decentralized Gradient Perspective

SKIM

Yuhang Deng, Zheng Chen, Erik G. Larsson

When consensus messages carry persistent additive noise, the authors add a decaying pull back toward each agent's own starting value, and they prove this recovers the exact average almost surely instead of a random value near it.

PROBLEM Standard consensus with diminishing stepsizes under persistent noisy links does reach agreement, but on a random value whose variance never vanishes, not on the true initial average. The noise accumulates in the network sum, and existing fixes are scattered methods with no shared design logic.

RESULTS The abstract gives no numbers. The main result is an almost sure convergence theorem, and the eight page paper has two figures, which suggests only a small illustrative simulation with no quantitative comparison against baselines.

METHOD They recast average consensus as decentralized gradient descent and add a state anchoring term, with a decaying weight, that keeps pulling each agent toward its initial value. The link noise is modeled as a martingale difference sequence with uniformly bounded conditional second moments. Under summability conditions linking the consensus stepsize to the anchoring weight, they prove almost sure convergence to the exact average and show that several earlier methods are special cases of the same gradient view.

CAVEAT The guarantee is purely asymptotic, and the abstract says nothing about convergence rate, even though a decaying anchor on top of a diminishing stepsize is likely to be very slow in practice. The noise model covers additive zero mean perturbations only, not packet loss, delay, time varying topology or agents that drop out. Those are exactly the failure modes that matter for vehicles. The unifying framework claim also needs checking against how much it really adds beyond known bias corrected and compensated noisy consensus schemes.

WHY IT MATTERS This is peripheral to your work. It is a pure consensus theory result on a fixed network with abstract link noise, and it has no maneuver, trajectory or vehicle model. It becomes relevant only if you build consensus based distributed trajectory planning and need an exact agreed quantity, such as a shared merge time or gap position, over noisy V2X links. Even then, the missing rate analysis and the lack of any packet loss model limit what you can take from it.

arXiv:2609.28082v1 PDF incremental

Action-Directed Information for Distributed Control and Agentic Interaction

SKIM

Shlomo Dubnov

A single-author study that uses a simulated four-legged walker to ask whether one limb reading its peers' force sensors helps control, and finds that the answer depends on where you measure the information.

PROBLEM In distributed control, it is unclear how to measure whether a message exchanged between components actually matters to the shared task. Standard quantities like Transfer Entropy and Directed Information are hard to estimate, and they are not tied to function under failure.

RESULTS The abstract gives no numbers. It says only that the peer-sensor controller has lower late tracking error in several conditions, and that the action-predictive estimator shows a substantially larger peer-message gain under compound failure. A second estimator, which predicts a future functional scalar, shows no stable advantage.

METHOD The author builds DI-Walker, a two-dimensional four-limb simulated body driven by frozen Cross-Entropy-Method policies. It compares a controller that uses each limb's own force sensor against one that uses the other limbs' force sensors. Information is measured at the receiver's action, using a finite-history estimator that predicts that action, and is linked to function through limb loss, limb slip and dropout of weak central control.

CAVEAT Everything rests on one toy simulated walker with frozen policies and no quantitative results in the abstract, and 'several conditions' suggests the benefit is not consistent. The two estimators disagree, and the paper presents this as a methodological lesson rather than resolving it. The author also openly concedes that the measures are not exact Directed Information and that no data-rate theorem is proven.

WHY IT MATTERS This is peripheral to your work: there are no vehicles, no negotiation, no proposal and acceptance, and no packet loss model. The one idea worth borrowing is to judge a maneuver message by how much it changes the receiver's action rather than the final outcome. That could inform how you evaluate intent-sharing formats under partial participation, but you can take that from the abstract alone.

arXiv:2609.27580v1 PDF incremental

Gregorio Marchesini et al.

Several space robots push a passive floating object around a single circular obstacle while a human steers it, using a centralized mixed-integer allocator with Lyapunov and barrier constraints that decides which robot pushes where, and it is tested only in Gazebo.

PROBLEM The robots can only push, never pull, so a given robot can produce force only in certain directions and only from where it is touching the target. When the needed push changes, robots have to move to new contact points, and during that time control over the target is only partial. Standard thrust allocation and safety filters assume continuous full actuation, so they don't cover this case.

RESULTS The abstract gives no numbers at all. Validation is Gazebo simulation only, with a single circular obstacle, and the abstract names no baseline, no team size, no delay values and no tracking error figures.

METHOD A central controller converts the human's commanded twist into pushing forces using a mixed-integer thrust allocation program. A control Lyapunov constraint handles tracking and a control barrier constraint keeps the target clear of the obstacle. The integer variables select which contact points are active, and the controller explicitly accounts for delay while robots move to new contact points.

CAVEAT Every result is simulated, the safety guarantee covers only one circular obstacle, and the allocation runs as one centralized mixed-integer program, so it is unclear whether it scales or runs in real time as the team grows. The chasers here don't negotiate or exchange intentions. A single optimizer assigns contacts, so the multi-agent part is really centralized allocation.

WHY IT MATTERS This is peripheral to your work. The robots don't negotiate, share messages or resolve conflicts with each other, and there is no model of message loss or partial participation. The only thread worth taking is how they keep barrier-function safety guarantees when control authority is intermittent and delayed. That might inform how you think about safety while a platoon is rearranging or a merge is still being agreed, but it doesn't transfer directly.

arXiv:2609.24376v1

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